CSE 374
Algorithms I

Chapter 0: Introduction

Instructor

Xianglong Feng

- Email: fengx17@miamioh.edu
- Office: DSB 374
- Office Hours:
  - Monday: 3:00 -6:00 PM

- Research Interests:
  - Hardware Security
  - Security of machine learning & machine learning for security
  - AR (hardware & software)
  - If you are interested, please send me an email with your CV.

Senior Capstone Projects

- https://sites.miamioh.edu/cse-senior-design-expo/2024/04/security-of-machine-learning-i/
- https://sites.miamioh.edu/cse-senior-design-expo/2024/04/mixed-meta-machine/
- https://sites.miamioh.edu/cse-senior-design-expo/2024/04/security-of-machine-learning-ii/
- https://sites.miamioh.edu/cse-senior-design-expo/2024/04/metaverse-interaction/
- https://sites.miamioh.edu/cse-senior-design-expo/2024/04/computer-vision-for-healthcare/
- https://sites.miamioh.edu/cse-senior-design-expo/2023/04/aifps-game-defense-2023-team1/
- https://sites.miamioh.edu/cse-senior-design-expo/2023/04/cellphone-arvr/
- https://sites.miamioh.edu/cse-senior-design-expo/2023/04/aifps-game-defense-2023-team2/
- https://sites.miamioh.edu/cse-senior-design-expo/2022/04/ar-for-hearing-loss/
- https://sites.miamioh.edu/cse-senior-design-expo/2022/04/ml-app-security/

TA

- Graduate TA:
    - Fiza Ashraf
    - Email: ashraff@miamioh.edu
    - Office Hours:
        - Wednesday 3-5PM
        - Thursday 1-2PM

Important time & date

- Class meets:
  - Section A:
    - Tuesday & Thursday
    - Section A: 2:50 PM -4:10PM
  - Section B:
    - Tuesday & Thursday
    - Section B: 1:15 PM -2:35PM
  - Section C:
    - Monday & Wednesday
    - Section C: 1:15 PM -2:35PM

Important time & date

- Important date:
  - Last day to Add/Drop through Banner Web: Wed Aug 28
  - Last Day for Departments to Add Students to a Course or Independent Work: Fri, Aug 30
  - Last day to drop without a grade: Friday Sep 13
  - Midterm Exam: (TBD)
  - Last day to drop with grade of W assigned: Mon Oct 28
  - Final Exam: (TBD)

Course Overview

- **Introduction to Algorithms**
  - Definition and importance of algorithms
  - Problem-solving and algorithmic thinking
  - Time and space complexity (Big O, Big Theta, Big Omega notation)
- **Basic Data Structures**
  - Arrays, Linked Lists, Stacks, and Queues
  - Trees (Binary Trees, Binary Search Trees, AVL Trees)
  - Hash Tables and Hash Functions
  - Heaps and Priority Queues
  - Graphs (Representation, Adjacency List/Matrix)

Course Overview

- **Sorting and Searching Algorithms**
  - Basic Sorting: Bubble Sort, Insertion Sort, Selection Sort
  - Advanced Sorting: Merge Sort, Quick Sort, Heap Sort
  - Searching: Binary Search, Linear Search
  - Order statistics (K-th smallest/largest element)
- **Divide and Conquer**
  - Concepts of Divide and Conquer
  - Merge Sort, Quick Sort, Binary Search, Strassen's Algorithm

Course Overview

- **Advanced Topics**
  - Graph Algorithm
  - Dynamic Programming
  - Greedy Algorithms
  - NP-Completeness and Computational Complexity

Course Materials

- https://algs4.cs.princeton.edu/home/
- https://leetcode.com/

Homework

- Homework:
  - Types:
    - Paper based:
      - PDF version (scan)
    - Programing:
      - Submit the code
  - Release time & due time:
    - Release: Thursday or Friday
    - Due: Friday of next XX week (No late submission!!!)
  - Help session:
    - Office hours or by appointment
    - Not available in Friday

Selected homework (coding)

- Sort
- Quicksort
- 2Sum from Leetcode
- BFS, DFS

How to finish

- Programming Language:
  - Java
- Platform:
  - Your PC
  - Submit to the Canvas

Quiz + in-class activity

Quiz + attendance (15%+15%)

- Bring your laptop in every class
- fully charge it before you come

Bonus (up to 4 points to the final grade, 0.5 pts for each)

- In class small quiz (with bonus):
  - Kahoot (top three)
- In class activity
  - The first correct answer
  - The one first raise hand

Midterm (Two times)

- One cheat-sheet:
  - A4 size
  - Handwritten
  - One side
  - Attach it to the exam

Final (One)

- One cheat-sheet:
  - A4 size
  - Handwritten
  - One side
  - Attach it to the exam

Final & Midterm

- Coding questions:
  - 80% of the coding questions are from the homework.
  - 20% of the questions are new.
- Q&A questions:
  - Based on the text materials.

Grading policy

- **Attendance (15%):** Attendance will be checked in the form of in-class quiz (non-graded)
- **Homework (30%)**: About 10 homework.
- **Quizzes (15%):** Cover the materials of the courses between the two quizzes. (graded)
- **Midterm Exam (20%)**:
- **Final Exam (20%):**
- **Extra Bonus (max10) to the final grade (not the final exam)**

Grading Scales

A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
>= 96.67	>= 93.33	>= 90	>= 86.67	>= 83.33	>= 80	>= 76.67	>= 73.33	>= 70	>= 66.67	>= 63.33	>= 60	< 60

Your full score: 110

Support

- Miami University is committed to ensuring equal access to students with disabilities. Miami's Office of Student Disability Services (SDS) assists students with determining eligibility for services and accommodation planning. Students who are entitled to disability-related academic adjustments, auxiliary aids, etc., must receive accommodations in university courses "https://miamioh.edu/student-life/sds/student-tools/how-to-register/index.html. Please understand that formal communication from SDS must be presented prior to the coordination of accommodations for this course. For more information, students may contact SDS at sds@miamioh.edu

sds@miamioh.edu

https://miamioh.edu/student-life/sds/student-tools/how-to-register/index.html

Resources and Support for Students

- As an instructor, I have a duty to report. This means I am required to promptly report to the Deputy Title IX Coordinator (titleix@miamioh.edu) any information a student shares with me regarding harassment, discrimination, sexual misconduct and interpersonal violence, or retaliation. A report does not initiate an investigation. It engages a discussion of your resources, supportive measures, and options available. If students want to speak with someone confidentially, the following resources are available on and off campus:
- Student Health Services, (513) 529-3000
- Student Counseling Services, (513) 529-4634

Resources and Support for Students

- Women Helping Women (WHW) Sexual and Interpersonal Violence Support Specialists are available to support all students and can be contacted by emailing mailto:mu@womenhelpingwomen.org. As well as calling/texting 513-846-8402 between 9AM-5PM. The 24-hour hotline is 513-381-5610. WHW supports ALL survivors of dating/domestic violence, sexual assault, and stalking, regardless of gender identity or sexual orientation.
- Speaking with a confidential resource person does not preclude students from making a formal report to the University if and when they are ready.

https://miamioh.edu/diversity-inclusion/programs-resources/report-incident/index.html

For more information, please visit https://miamioh.edu/campus-safety/sexual-assault/

and https://www.miamioh.edu/diversity-inclusion/oeeo/index.html.

Academic Integrity

Academic Integrity

Homework projects are intended to be individual assignments. This does not mean that you cannot discuss the problems at a "high level", but it does mean that you should not share code. Here are some guidelines that will help you:

- Assignments should be solved using the specific techniques and algorithms you have been learning in class. Don't search for solutions online because: you won't learn the right concepts, your code will likely not work as specified, and this is considered a violation of the Academic Integrity Policy.
- Don't share your code with anyone on homework projects. Don't ask for someone else to share their code. Attempts to copy code and then change it to look slightly different nearly always fail, and lead to students being found responsible for a violation of the Academic Integrity Policy.
- Don't post assignments or solutions to sites like Chegg or CourseHero because that is considered a violation of this course's Academic Integrity Policy. Plus, did you know that Miami's Academic Integrity office has full access to the materials on such sites?
- Don't reuse your old work or someone else's old work for this class, because doing so is considered a violation of the Academic Integrity Policy.
- In the CSE department, the default penalty for a first-time violation of the Academic Integrity policy is: a score of zero for the assignment/exam/quiz and a reduction in the student's final grade of a full letter grade (for example, a C- would become a D-).

However...you should work together! You should discuss problems, draw pictures, talk about algorithms, and so on. But that doesn't mean sharing code.

Generative AI

- Use of Generative AI is allowed in this course under the parameters outlined below. Generative AI is a tool that is important for you to know how to use. Like any tool, it has appropriate and inappropriate uses. AI also has limitations.
  - AI output is not always accurate, may be contradictory, may be biased, may be overly vague, and may not be current. AI has been known to “make stuff up,” particularly quotes and sources. You are responsible for what you submit. If you submit fake sources/citations/quotes, that will be academic dishonesty. If you submit inaccurate or unsubstantiated information, that will result in a poor grade.
  - I recommend that you find a credible source for any AI-provided information that you choose to use (the Librarians can help you with this!). I do not recommend asking AI to provide you with a list of sources.
  - The information generated by AI is limited by the prompt you provide.

Generative AI

- You may use AI to assist in idea/topic generation, organizing/outlining, creating early drafts, or polishing a final draft. Any use of AI in your work must be cited appropriately.
  - You are required to submit screenshots of your prompt and output.
  - You must indicate within the text when information from AI is being used and provide an appropriate reference section citation.
  - For some assignments, you may be required to annotate your assignment explaining your choices regarding the AI output.

Generative AI

- You are responsible for your final submission
  - Your submission should demonstrate your understanding of the information.
  - Your submission must meet the assignment guidelines.
  - You may **not** use AI to wholly generate your final submission.
  - Suspected use of AI to generate an entire submission or without appropriate citation/screenshot history will be reported as academic dishonesty.
- Any assignments on which AI CANNOT be used are noted in the syllabus.

Generative AI

- If more than one submissions share the same code block, the submissions will be graded with 0.